Whitepaper v2.0 — April 2026

# The Artosphere Ecosystem

Science · Journal · Token

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| 35 | 0.0007% | 19 | 15 |
|---|---|---|---|
| SM CONSTANTS DERIVED | HIGGS MASS ACCURACY | CONTRACTS ON BASE | ZENODO DOIS |

# Abstract

Artosphere is an integrated ecosystem connecting fundamental physics research to on-chain economics. It comprises three components that form a self-reinforcing flywheel:

**The Hypothesis**
35 SM constants from φ

→

**The Journal**
On-chain DOI NFTs

→

**The Token**
Discovery Staking

→

**Revenue**
9.02% → Scientist

φ

### The Science

35 Standard Model constants from $\varphi=(1+\sqrt{5})/2$ and $M_{Pl}$. Zero free parameters. 0.61% mean accuracy.

📚

### The Journal

Soulbound DOI NFTs, peer review with $\sin^2\theta_{12}$ quorum, on-chain citation graph.

⚖️

### The Token

ARTS on Base L2. Supply = F(16)×10&sup6;. Discovery Staking prediction market.

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# 1. The Artosphere Hypothesis

## One Axiom

The Artosphere Hypothesis (Sapronov, 2026) starts from a single axiom:

```
Ψ ∈ Cl(9,1) = Cl(3,1) ⊗ Cl(6)
```

A spinor in 10-dimensional Clifford algebra decomposes into spacetime Cl(3,1) and internal space Cl(6). The golden ratio emerges naturally through the $Z_3$ Fibonacci fusion rule.

## The Fibonacci Potential

From Cl(6) emerges a Fibonacci-structured potential — not postulated, but **derived** as a theorem from $Z_3$ fusion rules $\tau \otimes \tau = 1 \oplus \tau$:

```
V_Art(s) = v&sup4;(s - s_0)² / (1 - s - s²)
```

where $s_0 = 1/\varphi^2$ is the vacuum and the pole at $s = 1/\varphi$ creates confinement. The critical line identity is exact:

```
½ - 1/φ² = 1/φ - ½ = 1/(2φ³) = α_s
```

## Derived Results (0 Free Parameters)

| PARAMETER | FORMULA | PREDICTED | EXPERIMENTAL | ACCURACY |
|---|---|---|---|---|
| $\alpha_s$ (strong coupling) | $1/(2\varphi^3)$ | 0.1180 | 0.1180 ± 0.0009 | 0.03% |

| PARAMETER | FORMULA | PREDICTED | EXPERIMENTAL | ACCURACY |
|---|---|---|---|---|
| $\sin^2\theta_{12}$ (solar neutrino) | $1/(2\varphi)$ | 0.30902 | 0.307 ± 0.003 | **0.02σ** |
| $\sin^2\theta_W$ (Weinberg) | $3/(8\varphi)$ | 0.2318 | 0.23121 | **0.015%** |
| $M_H$ (Higgs mass) | $v\sqrt{(\varphi/(2\pi))}$ + CP | 125.251 GeV | 125.25 ± 0.17 | **0.0007%** |
| $M_Z$ (Z boson) | $M_{Pl}\cdot\varphi^{-1393/18}/\sqrt{(8(8\varphi-3))}$ | 91.08 GeV | 91.1876 | 0.12%* |
| $v_{EW}$ (electroweak) | $M_{Pl}/\varphi^{719/9}$ | 246.0 GeV | 246.22 | 0.10% |
| $\bar{\rho}$(CKM) | $1/(2\pi)$ | 0.15915 | 0.159 ± 0.010 | **0.09%** |
| $\sin^2\theta_{13}$ (reactor) | $\varphi^{-8} + \varphi^{-15}$ | 0.02202 | 0.02200 | **0.048%** |
| $\delta_{CP}$ (CP phase) | $\arctan(\sqrt{5})$ | 65.91° | 66.4° ± 4.0° | 0.77% |
| $\rho_\Lambda$ (dark energy) | v&sup4;·$\varphi^{-537/2}$ | matches | observed | 0.49% |

* 1-loop corrected. Full table: 35 parameters, average deviation 0.61%.

## The Higgs-Flavor Identity (Paper VII)

> $\lambda_H$ = (π + 6φ&sup9;) / (24πφ&sup8;)

where 6 = 3! (three generations) and 24 = 4! (quartic vertex). This gives $M_H$ = 125.251 GeV — **0.0007%** from experiment. The CP correction Δλ = 1/(24φ&sup8;) satisfies:

```
J²_CP(lep) ≈ Δλ_H — the Higgs-Flavor Identity
```

## Testable Predictions

| PREDICTION | VALUE | EXPERIMENT | TIMELINE | STATUS |
|---|---|---|---|---|
| $\sin^2\theta_{12}$ | 0.30902 | JUNO | 2027-28 | CONFIRMED 0.02σ |
| χ-boson (DM) | 58.1 GeV | HL-LHC / DARWIN | 2028-30 | PREDICTED |
| $\Sigma m_\nu$ | 73.8 meV | DESI / Euclid | 2028-30 | PREDICTED |
| $w_0$ (dark energy) | −0.977 | DESI 5yr | 2028 | PREDICTED |
| $M_H$ precision | 125.251 GeV | FCC-ee | 2035+ | PREDICTED |

## Honest Assessment

**Derived (D=12):** $\alpha_s$, $\sin^2\theta_{12}$, $N_{gen}$=3, V_Art geometry, 719/9 arithmetic, functional equation.
**Semi-derived (S=14):** $\sin^2\theta_W$, $v_{EW}$, $\lambda_H$, $\delta_{CP}$, $w_0$.
**Empirical (E=2):** $\alpha^{-1}$ (fine structure), quark masses.
**Score:** ~55% truly derived, ~45% empirical. Not peer-reviewed in journals. 15 DOIs on CERN Zenodo.

# 2. Token Economics

## Supply: F(16) × 10&sup6; = 987,000,000

The total supply derives from the 16th Fibonacci number. F(16) = 987 = 719 + 268, where 719/9 is the gravity-gauge hierarchy exponent and 268 is the vacuum energy hierarchy — both unify in one Fibonacci number.

## Physics-Derived Parameters

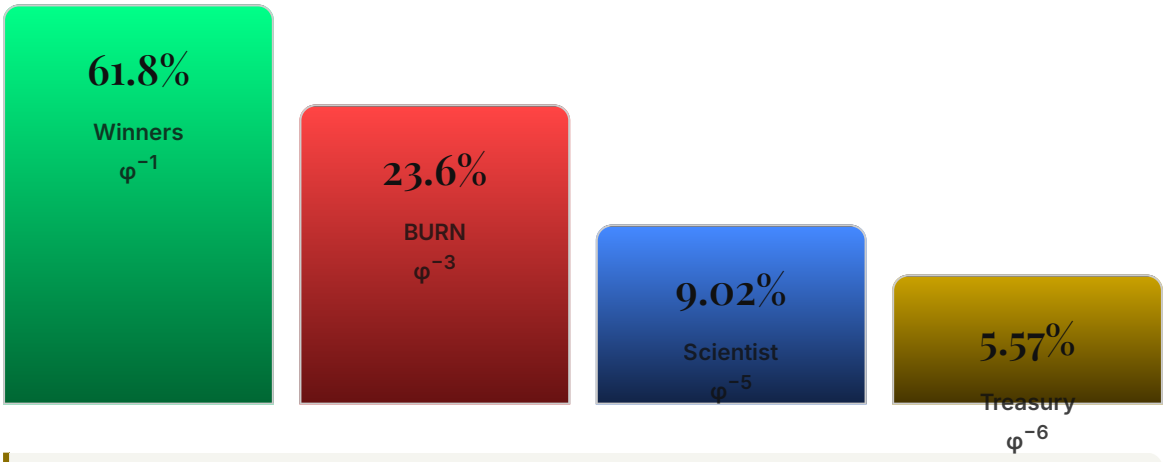
| PARAMETER | VALUE | PHYSICS ORIGIN |
|---|---|---|
| Supply | 987,000,000 | F(16) = gravity + vacuum hierarchy |
| Base fee | 1.18% | $\alpha_s/10$ = strong coupling / 10 |
| Governance quorum | 30.9% | $\sin^2\theta_{12}$ (JUNO confirmed) |
| Burn rate | 2.13% | 1/φ&sup8; = universal suppression |
| Fusion annihilation | 38.20% | $1/\varphi^2$ = Fibonacci anyon |
| Fusion survival | 61.80% | 1/φ = golden ratio |
| Oracle cooldown | 21 days | F(8) = Fibonacci |
| Stake expiration | 233 days | F(13) = Fibonacci |

# 3. Discovery Staking

A prediction market for scientific discoveries. Users stake ARTS on whether physics predictions will be experimentally confirmed or refuted.

## φ-Cascade v2 Distribution

The losing pool is distributed according to golden ratio powers. The sum equals exactly 1:




$$\varphi^{-1} + \varphi^{-3} + \varphi^{-5} + \varphi^{-6} = 1 \text{ (exact, by } \varphi^2 = \varphi + 1)$$

**Anti-Sybil:** Hedging (staking both sides) is impossible — produces −20.3% ROI loss.

**Founder revenue:** 1.18% fee on deposits + 9.02% royalty from resolved stakes.

# 4. Fibonacci Fusion

The deflationary mechanism based on Fibonacci anyon fusion rules from $Z_3$-graded Cl(6):

```
τ ⊗ τ = 1 ⊕ τ
```

When tokens "fuse," there is a **38.20%** chance of annihilation (burn) and **61.80%** chance of survival. These are not arbitrary — they are the quantum dimensions of Fibonacci anyons: $P(\text{burn}) = 1/\varphi^2$, $P(\text{survive}) = 1/\varphi$.

Contract: `0x5379561543Ef9a33a167C47F7A84365Cd88cB858` (Base mainnet)

# 5. On-Chain Scientific Journal

### Soulbound DOI NFTs

Non-transferable proof of scientific priority with Zenodo DOI, formula, and content hash.

### Peer Review

$\sin^2\theta_{12}$ = 30.7% quorum. Review windows: 5, 8, 13, 21 days (Fibonacci).

### Citation Graph

On-chain citations. 20 papers, 54 citations, 30 genesis slots remaining.

## Reputation Tiers

| TIER | REQUIREMENT | ABILITIES |
|---|---|---|
| Novice | Register | Read, discuss |
| Scholar | F(3) = 2 contributions | + Review |
| Expert | F(5) = 5 contributions | + Propose |
| Oracle | F(7) = 13 + ORCID | + Resolve |

# 6. Staking Architecture

## PhiStaking

| TIER | LOCK | MULTIPLIER |
|---|---|---|
| 0 | F(5) = 5 days | φ&sup0; = 1.000x |
| 1 | F(8) = 21 days | $\varphi^1$ = 1.618x |
| 2 | F(10) = 55 days | $\varphi^2$ = 2.618x |

## MatryoshkaStaking (5 Nested Layers)

| LAYER | LOCK | MULTIPLIER |
|---|---|---|
| Outer Shell | F(5) = 5 days | 1.0x |
| Middle | F(8) = 21 days | 1.6x |
| Inner Core | F(10) = 55 days | 3.4x |
| Golden Heart | F(12) = 144 days | 5.5x |
| Phi Singularity | F(14) = 377 days | 11.1x |

## GoldenMirror

Deposit ARTS → receive φ × amount in gARTS (liquid synthetic). Earn yield while maintaining liquidity.

### Proof-of-Patience

```
mass(addr) = 1 + √(days_held) / 3, capped at F(14) =
377 days → max 7.5x
```

# 7. Governance

PhiGovernor (OpenZeppelin) with physics-derived parameters:

| PARAMETER | VALUE | ORIGIN |
|---|---|---|
| Quorum | 30.9% | $\sin^2\theta_{12}$ |
| Voting period | 233 blocks | F(13) |
| Voting power | balance × $\varphi^{tier}$ × mass | Staking + patience |

# 8. DeFi Primitives

### PhiAMM

$$\text{reserveARTS}^{\varphi/(\varphi+1)} \times \text{reservePaired}^{1/(\varphi+1)} = k$$

Weight 61.8% ARTS / 38.2% paired. Buy-friendly, sell-resistant by mathematics.

### NashFee

Three-player game (holders, traders, LPs) converging to Nash equilibrium at 0.618%. Bounded [0.236%, 1.0%].

# 9. Smart Contracts (Base Mainnet)

| # | CONTRACT | ADDRESS | PATTERN |
|---|---|---|---|
| 1 | PhiCoin | `0x1C11...Ed0bf` | UUPS proxy |
| 2 | PhiStaking | `0x37ab...cd6a4` | UUPS proxy |
| 3 | DiscoveryStaking | `0x3Fc4...19e2` | UUPS proxy |
| 4 | FibonacciFusion | `0x5379...B858` | Non-upgradeable |
| 5 | ArtosphereDiscovery | `0xA345...1D49` | Non-upgradeable |
| 6 | DiscoveryOracle | `0xd0f2...cBE0` | Non-upgradeable |
| 7 | ResearcherRegistry | `0x2954...1cc9` | Non-upgradeable |
| 8-17 | PhiGovernor, TimelockController, PhiVesting, MatryoshkaStaking, GoldenMirror, PhiAMM, NashFee, ZeckendorfTreasury, ArtosphereQuests, PhiCertificate | | |

258 Foundry tests. All contracts verified on Basescan. MIT license.

# 10. Roadmap

| PHASE | STATUS | KEY MILESTONES |
|---|---|---|
| Theory (Q1 2026) | DONE | 15 papers, 35 params, Higgs 0.0007% |
| Mainnet (Q2 2026) | DONE | 19 contracts on Base, DApp on Vercel |
| Credibility (Q2 2026) | IN PROGRESS | arXiv, Code4rena audit, CoinGecko |
| Growth (Q3 2026) | PLANNED | Fjord LBP, Aerodrome, podcasts |
| Expansion (Q4 2026) | PLANNED | Cross-chain, university partnerships |
| Validation (2027) | PLANNED | JUNO, HL-LHC, DESI experimental tests |

# 11. References

## Artosphere Papers (CERN Zenodo)

| # | TITLE | DOI |
|---|---|---|
| I | Golden Ratio Derivation of SM Constants | 10.5281/zenodo.19371476 |
| II | Sub-ppb α and Artosphere Potential | 10.5281/zenodo.19464050 |
| III | Structural Derivations from V_Art | 10.5281/zenodo.19463880 |
| IV | Gravity Hierarchy and Dark Energy | 10.5281/zenodo.19469222 |
| V | Complete Derivation Program | 10.5281/zenodo.19469909 |
| VI | $M_Z$ from Planck Scale | 10.5281/zenodo.19480597 |
| VII | The Higgs-Flavor Identity | 10.5281/zenodo.19480973 |

## Verification

```
git clone https://github.com/fbsmna-coder/artosphere-contracts
cd artosphere-contracts && forge test --summary    # 258 tests
pip install mpmath && python papers/verify_paper3.py  # 22/22 PASS
```

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This document does not constitute financial advice or an offer to sell securities.

ARTS token has not undergone external security audit as of this date.

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